

Assignment

Q1. Make a redux app to make a counter which will start from 0 and 2 buttons with increment and decrement options and count will increase and decrease on the basis of which button.

Ans. Basic objectives :-

- ① Create a Redux-based counter app:
 - ① Initial app: 0 (value)
 - ② Two buttons:
 - Increment (+)
 - decrement (-)
 - ③ Count updates based upon button clicks.

Concepts used :-

- ① Redux Toolkit (@reduxjs/toolkit)
- ② React - Redux (useSelector, useDispatch)
- ③ Centralized state management

Steps to make the app :-

① Install Dependencies

npm install @reduxjs/toolkit react-redux

② Create Redux Store

```
import { configureStore } from '@reduxjs/toolkit';  
import counterReducer from '../features/counter/counterSlice';  
export const store = configureStore({
```

```

    reducer: {
      counter: counterReducer,
    },
  });

```

③ Create Counter Slice

```

import { createSlice } from '@reduxjs/toolkit';
const initialState = {
  value: 0;
};

```

```

const counterSlice = createSlice({
  name: 'counter',
  initialState,
  reducer: {
    increment: (state) => {
      state.value += 1;
    },
    decrement: (state) => {
      state.value -= 1;
    },
  },
});

```

```

export const { increment, decrement } = counterSlice.actions;
export default counterSlice.reducer;

```

④ Create Counter Component

```

import React from 'react';
import { useSelector, useDispatch } from 'react-redux';
import { increment, decrement } from './counterSlice';

```


Date _____
Page _____

```

const Counter = () => {
  const count = useSelector((state) => state.counter.value);
  const dispatch = useDispatch();
  return (
    <div style={{ textAlign: 'center' }}>
      <h1> Counter: {count} </h1>
      <button onClick={() => dispatch(increment())}> </button>
    </div>
  );
};
export default Counter;

```

⑤ Connect Redux to App

```

import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import { Provider } from 'react-redux';
ReactDOM.createRoot(document.getElementById('root')).render(
  <Provider store={store}>
    <App>
  </Provider>
);

```

⑥ Use counter in App

```

import Counter from './Counter';
function App() {
  return (
    <div>
      <Counter />
    </div>
  );
};

```

Diagram 1-

